EXECUTIVE SUMMARY

E-Commerce Sales Analysis Using Python

Project Overview

This project focuses on analyzing e-commerce sales data to understand business performance, identify profitability patterns, and develop data-driven recommendations. Python was used to clean, process, analyze, and visualize the dataset to extract meaningful business insights.

Objectives

The main objectives of this project are to evaluate sales and profit performance across regions and product categories, identify the customer segment generating the highest revenue, examine the relationship between discounts and profitability, analyze the impact of shipping costs on profit margins, and recommend strategies to improve business performance.

Tools and Technologies

- **Python**: Core programming and data analysis
- **Pandas and NumPy:** Data cleaning, manipulation, and calculations
- **Matplotlib and Seaborn**: Data visualization and exploratory analysis
- **Jupyter Notebook:** Executing code and documenting results

Methodology

The analysis began with data inspection and cleaning, including checking missing values, removing exact duplicate records, correcting data types, validating dates, standardizing text fields, and investigating unusual values. Exploratory data analysis was then performed using summary statistics, charts, grouped comparisons, and correlation analysis. The results were used to

answer six key business questions and develop actionable recommendations.

Key Business Findings

The analysis examined six major areas:

1. **Regional profitability:** Identified the region with the highest total profit.
2. **Product category performance:** Compared sales, total profit, and average profit to identify promising categories for promotion.
3. **Discount effectiveness:** Examined average profit across discount bands and the relationship between discount and profit.
4. **Customer segment revenue:** Identified the segment generating the highest total sales.
5. **Shipping costs:** Investigated the relationship between shipping expenses, profit, and profit margins.
6. **Business recommendations:** Developed five recommendations based on the observed findings.

Recommendations

The project proposes five areas for business improvement: strengthen regional profitability, promote profitable product categories, optimize discount policies, improve customer retention, and investigate opportunities to control shipping expenses. These recommendations should be prioritized according to the actual findings.

Conclusion

This project demonstrates how Python-based data analysis can support business decision-making by transforming raw e-commerce data into useful insights. The results can help identify growth opportunities, understand profitability patterns, and guide improvements in marketing, pricing, customer management, and shipping operations.